

Working

1. Caesar Cipher

The Caesar cipher encrypts each letter by shifting it a fixed number of positions in the alphabet. The same shift is applied to every character. For example, with a key of **6**, H becomes N, E becomes K, and so on. Decryption reverses the same shift.

2. Vigenère Cipher

The Vigenère cipher uses a **keyword** to determine different shifts for different letters. The keyword is repeated until it matches the plaintext length. Each plaintext letter is shifted according to the corresponding key letter. Decryption uses the same key in reverse.

3. Rail Fence Cipher

The Rail Fence cipher is a **transposition cipher**, meaning it does not change the letters. Instead, the plaintext is written in a **zig-zag pattern** across a fixed number of rails and then read row by row to produce the ciphertext. Decryption reconstructs the zig-zag pattern to recover the original message.

Effectiveness Comparison

Cipher	Working	Security	Ease of Breaking	Effectiveness
Caesar	Fixed alphabetic shift	Low	Very easy	Low
Vigenère	Multiple shifts using a key	Moderate	Harder than Caesar	Better
Rail Fence	Rearranges letters using zig-zag pattern	Low	Relatively easy	Low

Conclusion: Among these three, **Vigenère provides better security**, while Caesar and Rail Fence are mainly useful for understanding basic cryptographic concepts. None of them is suitable for modern secure communication.